Observation-Based Outlier Detection in Categorical Data

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1 Introduction

In many practical applications, such as fraud detection, quality control, and market analysis, it is essential to identify anomalous patterns or outliers in data. Particularly in categorical datasets, unusual combinations of category values may indicate errors, special cases, or areas requiring further investigation. The observation-based outlier detection algorithm presented herein is designed to identify such anomalies by leveraging the frequency distribution of categorical variables.

The core idea behind the algorithm is to model the "normal" behavior of the dataset by constructing a contingency table that captures the joint frequencies of values across two or three categorical variables. By computing conditional proportions for these combinations, the method identifies those pairs (or triples) that occur far less frequently than expected relative to their marginal distributions. If a particular combination has a conditional proportion below a predefined threshold, it is flagged as potentially anomalous.

While the method is initially developed for two variables, it is readily extendable to three variables. The extension involves generalizing the contingency table to a higher dimension and computing multiple conditional proportions to capture the interactions among more variables. Nevertheless, as dimensionality increases, issues such as data sparsity, computational complexity, and challenges in visualization arise, making the two- and three-variable cases particularly attractive for practical applications.

2 Hypotheses of the Algorithm

The design and application of the observation-based outlier detection algorithm rely on several key assumptions:

- **Rarity Implies Anomaly:**

  Anomalous behavior is assumed to manifest as events or combinations of features that occur very infrequently. Rarity is quantified via conditional proportions derived from a contingency table. If a combination (e.g., (a, b) or (a, b, c)) has a conditional proportion significantly lower than a threshold T, it is likely an anomaly.

- **Coherent Normal Patterns:**

  It is assumed that most observations in the dataset conform to stable, coherent patterns (inliers). These normal patterns serve as a baseline for detecting deviations.

- **Limited-Dimensional Relationships:**

  The algorithm focuses on two- or three-dimensional interactions since these yield the most interpretable and statistically robust relationships. High-dimensional interactions face combinatorial explosion, increased data sparsity, and substantial challenges in visualization.

3 Baseline: The Two Variables Algorithm

This section presents the theoretical framework for detecting outliers in categorical data when considering two variables. The approach, largely inspired by a Volkan Yurtseven medium article [1] is based on frequency counts from a contingency table and conditional proportions that measure the typicality of a given combination.

3.1 Contingency Table Construction

Let A and B be two categorical variables. For any pair (a, b) with $a \in \mathcal{A}$ and $b \in \mathcal{B}$, define the frequency count as

$$f(a, b) = \text{number of occurrences where } A = a \text{ and } B = b.$$

The collection

$$\{f(a, b) \mid a \in \mathcal{A},\ b \in \mathcal{B}\}$$

forms a contingency table summarizing the joint distribution of A and B.

3.2 Conditional Proportions

Conditional proportions are computed as:

$$p(B = b \mid A = a) = \frac{f(a, b)}{\sum_{b' \in \mathcal{B}} f(a, b')} \triangleq \frac{f(a, b)}{f(a)}$$

and

$$p(A = a \mid B = b) = \frac{f(a, b)}{\sum_{a' \in \mathcal{A}} f(a', b)} \triangleq \frac{f(a, b)}{f(b)}.$$

These ratios help measure how typical or rare the pairing (a, b) is relative to the overall frequencies of A and B.

3.3 Outlier Detection Criterion

A key hypothesis is that if one or both conditional proportions are very low (i.e., below a threshold T), the combination (a, b) is anomalous. For example, setting $T = 0.01$ flags a pair as an outlier if:

$$p(B = b \mid A = a) < T \quad \text{or} \quad p(A = a \mid B = b) < T.$$

Example: Two-Column Case (Cold-Chain Food Distribution)

Consider a cold-chain food distribution company. Each record corresponds to a shipment that must respect temperature constraints. We track:

- **TEMPERATURE BAND** $A \in \{\text{Ambient}, \text{Chilled}, \text{Frozen}\}$,
- **HANDLING HUB TYPE** $B \in \{\text{AmbientHub}, \text{ColdHub}, \text{FrozenHub}\}$.

Assume the following simplified contingency table for (A, B):

HUB TYPE **TEMPERATURE BAND**	AmbientHub	ColdHub	FrozenHub	**Total**
Ambient	5200	180	8	5388
Chilled	40	4650	25	4715
Frozen	**6**	90	3650	3746
Total	5246	4920	3683	**13849**

Focusing on the combination $(\text{Frozen}, \text{AmbientHub})$:

- Combination count: $f(\text{Frozen}, \text{AmbientHub}) = 6$.
- Total Frozen counts: $f(\text{Frozen}) = 3746$.
- Hence, $p(\text{AmbientHub} \mid \text{Frozen}) = \frac{6}{3746} \approx 0.00160$.

With threshold $T = 0.01$, since $0.00160 < 0.01$, the pair $(\text{Frozen}, \text{AmbientHub})$ is flagged as an outlier. Operationally, this can indicate a misrouting event, an incorrect temperature classification, or a data entry issue that should be reviewed because it may imply a cold-chain compliance breach.

4 Extension to Three Variables

Suppose that we have three categorical variables A, B, and C. For a given combination of outcomes (a, b, c), we define the joint frequency as

$$f(a, b, c) \;=\; \text{number of occurrences of the event } \{A = a,\; B = b,\; C = c\},.$$

These frequencies are organized in a three-dimensional contingency table [2].

4.1 Conditional Proportions

To extend the algorithm to three variables, we compute conditional proportions by "summing out" one variable while holding the other two fixed. We define:

Conditioning on two variables: When handling three variables, we fix two of the variables at a time to gain information from this grouping, and look at the frequency of the last variable.

The marginal (conditional) sum over C is given by

$$f(a, b) = F\big(C \mid A = a, B = b\big) \;=\; \sum_{c' \in \mathcal{C}} f(a, b, c')\,.$$

Then, the conditional ratio (relative frequency) for the specific outcome $C = c$ is

$$p\big(C = c \mid A = a, B = b\big) \;=\; \frac{f(A = a, B = b, C = c)}{F\big(C \mid A = a, B = b\big)}\,.$$

Computing three ratios :

Therefore we compute:

$$p\big(C = c \mid A = a, B = b\big) \;=\; \frac{f(a, b, c)}{f(a, b)}\,,$$

$$p\big(B = b \mid A = a, C = c\big) \;=\; \frac{f(a, b, c)}{f(a, c)}\,.$$

and

$$p\big(A = a \mid B = b, C = c\big) \;=\; \frac{f(a, b, c)}{f(b, c)}\,.$$

4.2 Outlier Detection Criterion for Three Variables

A candidate combination (a, b, c) is flagged as an outlier if at least two out of the three conditional ratios fall below a predefined threshold T. In mathematical terms, (a, b, c) is considered an outlier if

$$\big\{p\big(C = c \mid A = a, B = b\big),\; p\big(B = b \mid A = a, C = c\big),\; p\big(A = a \mid B = b, C = c\big)\big\} < T$$

in at least two cases.

This extension allows the algorithm to capture multi-dimensional anomalies by considering the interactions among three variables. The criterion based on multiple conditional proportions improves robustness and reduces the risk of detecting spurious outliers.

Example: Three-Column Case (Supplier–Hub–Customer)

Consider a food distribution network where each shipment record includes:

- A: **SUPPLIER REGION** (e.g., `NorthEast`, `South`, `International`),
- B: **DISTRIBUTION HUB** (e.g., `Hub_A`, `Hub_B`, `Hub_C`),
- C: **CUSTOMER SEGMENT** (e.g., `Grocery`, `Restaurant`, `FoodBank`).

Suppose the triplet $(a, b, c) = (\texttt{International}, \texttt{Hub_A}, \texttt{FoodBank})$ occurs very rarely:

$$f(\texttt{International}, \texttt{Hub_A}, \texttt{FoodBank}) = 2.$$

However, the corresponding pairwise totals are substantial:

$$\begin{aligned} f(\text{International}, \text{Hub_A}) &= 900, \\ f(\text{International}, \text{FoodBank}) &= 500, \\ f(\text{Hub_A}, \text{FoodBank}) &= 700. \end{aligned}$$

The three conditional ratios become:

$$p(C = \texttt{FoodBank} \mid A = \texttt{International}, B = \texttt{Hub_A}) = \frac{2}{900} \approx 0.00222,$$

$$p(B = \texttt{Hub_A} \mid A = \texttt{International}, C = \texttt{FoodBank}) = \frac{2}{500} = 0.004,$$

$$p(A = \texttt{International} \mid B = \texttt{Hub_A}, C = \texttt{FoodBank}) = \frac{2}{700} \approx 0.00286.$$

With $T = 0.01$, at least two of these ratios fall below the threshold, so the triplet is flagged as an outlier. In practice, such a pattern may indicate a special-case routing workflow, a supplier region mapping error, or an incorrect customer segment label.

4.3 Interpretational Challenges

Extending to three variables increases computational complexity and sparsity. Moreover, visualizing these multi-dimensional interactions is more challenging, and the threshold T may require fine-tuning or even adaptive methods to retain both sensitivity and specificity.

4.4 Reduction to a single Variable

In some cases, the grouping yields a group with only one non-constant category. In this case one might be interested in still running the algorithm of a single variable.

The degenerated logic becomes:

We have one categorical variable A which gives an outcome a. $f(a)$ counts the number of occurrences.

The proportion becomes:

$$p(A = a) = \frac{f(a)}{\sum_{a'} f(a')} = \frac{f(a)}{f(\text{group})} \quad \text{and} \quad p(A = a) < T$$

5 Challenges and Strategies for High-Dimensional Combination Selection

When extending outlier detection beyond two or three categorical variables, the following issues arise:

- **Combinatorial Explosion:**
  For n variables, even low-dimensional (binary or triplet) combinations grow rapidly as $\binom{n}{2}$ or $\binom{n}{3}$.
- **Data Sparsity:**
  Higher-dimensional contingency tables tend to be sparse, leading to unreliable conditional proportion estimates.

- **Interpretability:**
  Analyzing and visualizing high-dimensional interactions is complex. Thus, the algorithm focuses on selecting the most informative two- or three-variable combinations.

5.1 Combination Selection Strategy

To address the challenges posed by high-dimensional categorical data and to maintain interpretability, we implement a multi-step strategy for selecting variable combinations. This procedure ensures that only the most robust, unique, and informative combinations are used in subsequent analyses. The strategy involves the following steps:

1. **Domain-Driven Pre-Selection:**
   Informed by expert knowledge from the banking domain, we begin by filtering the full set of candidate variables to retain only those that are known to be relevant. This domain-driven pre-selection dramatically reduces the number of variables and, consequently, the number of combinations to evaluate. It focuses our attention on features that are more likely to interact meaningfully, improving both computational efficiency and the clarity of the final analysis.

2. **Statistical Evaluation with Significance Testing:**
   For the variables retained after pre-selection, we systematically evaluate all possible pairwise and triplet combinations. This evaluation consists of two main components:

   - **Computation of Association Metrics:**
     For binary combinations (i.e., pairs of variables), we compute an association metric using Cramer's V, which quantifies the strength of the relationship on a scale from 0 (no association) to 1 (perfect association). For triplet combinations (i.e., groups of three variables), we extend this approach by calculating a chi-square based metric—yielding a ϕ coefficient—that captures the multidimensional interaction.
   - **Derivation of P-Values:**
     Alongside the association metrics, we perform chi-square tests to obtain a p-value for each combination. The p-value measures the proportion that the observed association could have occurred by chance. By adopting a predetermined significance threshold (typically $p < 0.05$), we ensure that each combination we consider for further analysis both exhibits a high association metric (i.e., a strong relationship) and has a p-value low enough to indicate that the relationship is statistically reliable.

3. **Redundancy Reduction:**
   In practice, many binary combinations appear as components of triplets, potentially leading to duplicated or overlapping information. To avoid redundancy, we compare the association metric of a binary pair (M_{AB}) with that of the corresponding triplet (M_{ABC}) using a margin criterion:

$$M_{AB} \geq M_{ABC} + \delta,$$

where δ is a predetermined margin that specifies the minimum additional strength required for the binary combination to be considered unique. Moreover, we verify that the binary's p-value is suitably more significant than the corresponding triplet's p-value. If the binary combination does not satisfy these criteria, it is deemed redundant and excluded from the final list.

$$p_{AB} < p_{ABC}$$

If a pair overshadows both of its triplet, then those triplets are deemed redundant since the correlation among them are carried by the pair. The triplets are dropped from the list of final contestants.

4. **Selection of Top, Non-Overlapping Combinations:**
Finally, from the remaining candidates, we select the top non-overlapping combinations that provide distinct and actionable insights. For example, if our candidate features include A, B, C, D, and E, our procedure may yield non-redundant groups such as (A, B, C) and (D, E) or alternatives like (A, C) and (B, E). The goal is to ensure that each selected set reflects a unique interaction pattern, avoiding duplication while encapsulating the strongest associations.

This comprehensive selection strategy combines domain insights with rigorous statistical evaluation and redundancy control. By retaining only those combinations that exhibit both strong association and statistical significance, we enhance the reliability and interpretability of the subsequent outlier detection process.

Example: High-Dimensional Combination Selection

Assume a food distribution company wants to detect anomalies in shipment routing and cold-chain handling. After domain-driven pre-selection, the following four categorical variables are retained:

`TEMPERATURE BAND`, `DISTRIBUTION HUB`, `TRANSPORT MODE`, `CUSTOMER SEGMENT`.

Intuitively, these variables should exhibit structured dependence (e.g., frozen shipments should preferentially use refrigerated transport and specific hubs).

Step 1: Compute Association Metrics and p-values

After computing association metrics for all possible combinations—using Cramer's V for binary pairs and a chi-square–based metric for triplets—we obtain the following illustrative results.

Binary Combinations (Cramer's V):

Binary Combination	V	p-value
(TEMPERATURE BAND, DISTRIBUTION HUB)	0.68	0.005
(TEMPERATURE BAND, TRANSPORT MODE)	0.78	0.002
(TEMPERATURE BAND, CUSTOMER SEGMENT)	0.45	0.030
(DISTRIBUTION HUB, TRANSPORT MODE)	0.60	0.010
(DISTRIBUTION HUB, CUSTOMER SEGMENT)	0.35	0.040
(TRANSPORT MODE, CUSTOMER SEGMENT)	0.55	0.015

Triplet Combinations (chi-square–based metric):

Triplet Combination	M	p-value
(TEMPERATURE BAND, TRANSPORT MODE, DISTRIBUTION HUB)	0.80	0.002
(TEMPERATURE BAND, TRANSPORT MODE, CUSTOMER SEGMENT)	0.62	0.012
(TEMPERATURE BAND, DISTRIBUTION HUB, CUSTOMER SEGMENT)	0.55	0.020
(DISTRIBUTION HUB, TRANSPORT MODE, CUSTOMER SEGMENT)	0.58	0.018

Step 2: Handle Redundancy in Combinations

Each binary pair AB is compared against every triplet ABC that contains it. A binary pair is deemed redundant relative to a triplet if it does not provide materially stronger evidence of association. Using a margin $\delta = 0.05$, we test:

$$M_{AB} \geq M_{ABC} + \delta \quad \text{and} \quad p_{AB} < p_{ABC}.$$

Example Comparison:

$$\texttt{(TEMPERATURE BAND, TRANSPORT MODE)} \subset \texttt{(TEMPERATURE BAND, TRANSPORT MODE, DISTRIBUTION HUB)}$$

with:

$$M_{\texttt{TEMP,TRANS}} = 0.78, \quad M_{\texttt{TEMP,TRANS,HUB}} = 0.80.$$

The margin check becomes:

$$0.78 \geq 0.80 + 0.05 \quad \Longrightarrow \quad 0.78 \geq 0.85,$$

which fails. Therefore, the binary pair (TEMPERATURE BAND, TRANSPORT MODE) is treated as redundant relative to the triplet (TEMPERATURE BAND, TRANSPORT MODE, DISTRIBUTION HUB).

After redundancy reduction, candidate groups may include:

- **Triplet:** (TEMPERATURE BAND, TRANSPORT MODE, DISTRIBUTION HUB) with $M = 0.80,\ p = 0.002,$

- **Triplet:** `(DISTRIBUTION HUB, TRANSPORT MODE, CUSTOMER SEGMENT)` with $M = 0.58,\ p = 0.018$,
- **Binary:** `(TEMPERATURE BAND, DISTRIBUTION HUB)` with $V = 0.68,\ p = 0.005$ (if not deemed redundant under the comparison rules).

Step 3: Final Selection of Non-Overlapping Groups

Since several candidate groups share variables, a final selection retains only a small set of non-overlapping combinations that capture distinct operational structure. For instance, the strongest triplet

`(TEMPERATURE BAND, TRANSPORT MODE, DISTRIBUTION HUB)`

can be retained to model cold-chain logistics constraints, while an additional independent combination (if available) can be retained to capture customer-specific routing effects.

Downstream usage: The retained combinations are then used as inputs to the conditional-proportion outlier detection procedure to flag rare and potentially problematic operational patterns (e.g., frozen shipments using ambient transport through a hub that rarely handles frozen goods).

6 Tunable Parameters

The performance of our outlier detection algorithm depends critically on several key parameters that govern how outliers are flagged, data clusters are validated, and redundancies are reduced. Careful tuning of these parameters is essential to balance sensitivity and robustness in the presence of noise.

6.1 Outlier Detection Algorithim Parameters

Outlier Threshold (T): This threshold serves as the primary criterion for flagging a variable combination as an outlier. In practice, a combination is marked as anomalous if its conditional proportion—such as $p(B = b \mid A = a)$ for a binary case—falls below T. To choose an appropriate value for T, several strategies can be used:

- **Empirical Analysis:** Evaluate historical data to determine the typical conditional proportions for known anomalies. For example, if most true anomalies occur below 1%, setting $T = 0.01$ is supported by the data.
- **Cross-Validation:** Partition the dataset to test different values of T and select the threshold that provides the best balance between detecting true anomalies and avoiding false positives.
- **Adaptive Thresholding:** Utilize dynamic methods, such as quantile-based thresholds or bootstrap confidence intervals, so that T adjusts automatically to variations in the data.

Redundancy Margin (δ): In the redundancy reduction step, the redundancy margin is used to decide whether a binary combination provides additional information relative to a corresponding triplet. Specifically, if a binary pair's association metric M_{AB} does not exceed that of the associated triplet M_{ABC} by at least δ, the binary combination is considered redundant and is removed:

$$M_{AB} \geq M_{ABC} + \delta.$$

Typically, δ is set between 0.05 and 0.10, ensuring that only uniquely strong associations are retained.

6.2 Local Outlier detection Parameters

Elimination of Unstable Combinations During Modeling: Although the combination determination process is performed before the modeling phase, we apply an additional filtering step during model training. In the fit phase, the algorithm first sorts the inlier clusters of a given combination from smallest to largest based on their population sizes. It then calculates the rate of change between successive clusters. When the ratio between adjacent cluster populations exceeds a predefined threshold—the `local_cluster_jumping_decision_point` (default value 10)—this signals a significant jump in frequency. At that juncture, we compute the ratio of the cumulative population of the clusters up to this jump relative to the total population of the combination. If this cumulative ratio exceeds a second threshold—the `local_cluster_ratio_decision_point` (default value 0.05)—we interpret this as an indication that too many local clusters are present, implying that the segmentation is not sufficiently distinct. In such cases, the combination is eliminated from further consideration. This extra filtering step ensures that the retained combinations tend to exhibit a Pareto-like distribution, characterized by a few large clusters and only a handful of local clusters, which is deemed more meaningful for categorical data analysis.

Example: Four-Variable Case with Local Cluster Filtering

Consider a candidate combination involving the variables `TEMPERATURE BAND`, `TRANSPORT MODE`, `DISTRIBUTION HUB`, and `CUSTOMER SEGMENT`. After the initial combination determination process (based on association metrics and significance testing), assume that in the training (fit) phase the algorithm identifies several inlier clusters for this combination. Suppose that after grouping similar observations, the clusters are sorted (in ascending order by population) and yield the following frequency counts:

$$\text{Cluster Populations: } [5,\ 7,\ 10,\ 15,\ 20,\ 250,\ 300].$$

The total population in the combination is:

$$5 + 7 + 10 + 15 + 20 + 250 + 300 = 607.$$

Next, we compute the ratio between consecutive cluster populations:

- From 5 to 7: $7/5 = 1.40$
- From 7 to 10: $10/7 \approx 1.43$

- From 10 to 15: $15/10 = 1.50$
- From 15 to 20: $20/15 \approx 1.33$
- From 20 to 250: $250/20 = 12.50$
- From 250 to 300: $300/250 = 1.20$

Here, the ratio of 12.50 (from 20 to 250) exceeds the `local_cluster_jumping_decision_point` (default value 10), indicating a significant jump in the frequency counts. Next, we determine the cumulative population of the smaller clusters (those before the jump). Summing the populations of the first five clusters gives:

$$5 + 7 + 10 + 15 + 20 = 57.$$

We then calculate the ratio of this cumulative population to the total population:

$$\text{Cumulative Ratio} = \frac{57}{607} \approx 0.0939 \ (9.39\%).$$

Since 9.39% is greater than the `local_cluster_ratio_decision_point` (default value 0.05 or 5%), we conclude that the number of local clusters is too high. In other words, rather than exhibiting a Pareto-like distribution—with a few large clusters and only a few small local clusters—the combination is overly fragmented. As a consequence, this candidate combination is eliminated from further analysis during the modeling phase.

This example illustrates how, even if a variable combination is initially promising, an additional filtering step—based on the rate of change in inlier cluster sizes—can help eliminate combinations that do not exhibit the desired robust structure. By discarding combinations with excessive local fragmentation, we ensure that only those combinations with a clear, dominant cluster structure (i.e., reflective of meaningful interactions) are retained for outlier detection.

Local Cluster Threshold: This parameter determines whether the frequency count of a particular combination is high enough to be considered statistically reliable. In datasets with large volumes, a higher threshold may be appropriate, whereas in sparser datasets, a lower threshold helps to avoid discarding useful information. This adjustment helps in filtering out combinations that might merely represent noise.

Local Cluster Jumping Decision Point and Ratio Decision Point: During the local cluster analysis, the jumping decision point is used to detect significant breaks in sorted frequency counts, identifying where clusters suddenly change. Correspondingly, the ratio decision point sets the acceptable level of fragmentation by comparing the cumulative count of small clusters to the total count. Tuning these parameters requires examining the frequency distribution: increasing the jumping decision point makes the method less sensitive to minor fluctuations, while the ratio decision point is adjusted to enforce stricter or looser

fragmentation criteria based on the overall distribution pattern.

6.3 Summary Table of Tunable Parameters:

For a quick reference, Table 1 below summarizes the key parameters, the recommended methods for tuning them, and some practical guidance.

Parameter	**Tuning Method**	**Practical Guidance**
Outlier Threshold (T)	Empirical analysis, cross-validation, adaptive methods	Set using historical data and sensitivity analysis (e.g., $T = 0.01$).
Local Cluster Threshold	Empirical volume-based analysis	Adjust based on overall data density and volume.
Local Cluster Jumping Decision	Statistical and sensitivity analysis	Tune by examining significant frequency jumps.
Local Cluster Ratio Decision Point	Distribution-based evaluation	Set based on acceptable levels of fragmentation.
Redundancy Margin (δ)	Statistical redundancy analysis	Typically set between 0.05 and 0.10 to avoid duplications.

Table 1: Summary of Tunable Parameters

7 Addressing Practical Challenges and Robustness

To ensure the algorithm is both robust and practically applicable, several strategies are incorporated to address common challenges such as imbalanced data, noise, and the need for domain expertise. First, imbalanced datasets can lead to zero-frequency issues when estimating conditional proportions. To mitigate this, smoothing techniques (e.g., Laplace smoothing) are applied, and Bayesian methods that incorporate prior distributions based on domain knowledge may be employed to improve probability estimates for rare categories. This adjustment is critical in ensuring that the algorithm does not erroneously flag rare but legitimate data points as anomalies.

Another challenge is distinguishing between noise and legitimate rare events. To address this, it is advisable to integrate domain-informed validation steps. For example, organizing expert review sessions can help verify whether flagged rare combinations are truly anomalous. Additionally, statistical filtering methods, such as permutation tests or standardized residual analysis, can be utilized to further differentiate genuine anomalies from mere noise. These approaches add a layer of statistical rigor to the process, ensuring that the algorithm remains both sensitive and specific.

Finally, the practical integration of domain expertise is essential for refining the detection process. Structured expert review sessions—using methods such as the Delphi technique or structured questionnaires—assist in identifying and pre-selecting the most relevant variables

for the analysis. Moreover, incorporating domain-informed weighting schemes ensures that key business or operational variables are given higher priority, thereby aligning the detection process more closely with real-world considerations. Together, these strategies enhance the overall robustness and reliability of the outlier detection algorithm.

8 Limitations

While the observation-based outlier detection algorithm is effective in identifying rare combinations in categorical data, it is subject to certain limitations:

- **Sensitivity to Threshold Selection:**
  The detection performance critically depends on choosing appropriate thresholds (e.g., `T`,

9 Conclusion

This article has introduced an observation-based outlier detection algorithm tailored for categorical data. The methodology is founded on constructing contingency tables and computing conditional proportion ratios to flag anomalous combinations.

A detailed example using the `DIVISION` and `LOCATION` columns illustrated how an outlier (e.g., (Fashion, LOC4)) is identified due to its rare occurrence. The extension to three variables has been presented with the necessary modifications in computing conditional proportions and adapting the outlier criterion.

Furthermore, the document discusses strategies for combination selection in high-dimensional data, introduces tunable parameters with explicit justification (including empirical, cross-validation, and adaptive methods), and outlines practical challenges with corresponding robustness strategies.

Future work could explore automated threshold selection and further improve the integration of domain expertise, ensuring the method adapts optimally to various practical applications.

This article is based on [1]–[3].

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